C.V. Raman Global University, Bhubaneswar
Odisha – 752054, India
Department of Computer Science and Engineering

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A Comparative Benchmark of CNN Architectures for Image Classification on CIFAR-10

with Adversarial Robustness Analysis

Report

SUBHAM NAYAK
Regd. No. – 2301020045

Under the supervision of
Dr. MAMATARANI DAS
Dept. of Computer Science & Engineering

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Abstract

This report benchmarks five CNN architectures on CIFAR-10, asking a practical question: how do pretrained models compare to a custom baseline when training conditions are held constant? I tested EfficientNet-B0, MobileNetV2, ResNet-18, and VGG-16 — all fine-tuned from ImageNet weights — against a SimpleCNN trained from scratch. Every model ran for 20 epochs, Adam optimizer, CosineAnnealingLR, same augmentation pipeline. No exceptions.

EfficientNet-B0 topped the board at 88.11% with 4.02M parameters. MobileNetV2 hit 87.63% with only 2.24M — the best accuracy-per-parameter of any pretrained model. ResNet-18 came in at 87.33%. VGG-16 failed outright: 31.06%, barely above random chance. The reason is straightforward and fixable — five max-pool layers collapse a 32×32 input to a 1×1 feature map. Nothing left to classify.

The hardest classification boundary in the whole dataset is cat vs. dog — 9–11% mutual confusion even in the best model. I also compared these results against Dingeto & Kim's (2024) adversarial robustness findings. Their regularized ViT (Elastic Net) holds 82.2% under adversarial attack, while EfficientNet-B0 hasn't been tested adversarially here. That gap matters for real deployment decisions.

Keywords: Convolutional Neural Networks, Image Classification, CIFAR-10, Transfer Learning, Benchmark Study, ResNet, VGG, MobileNet, EfficientNet, Adversarial Robustness, Vision Transformers.

1. Introduction

CNN architecture research since AlexNet (2012) has produced a clear progression: VGGNet showed depth works, ResNet showed skip connections fix vanishing gradients, MobileNetV2 showed you can get competitive accuracy in a fraction of the parameters, and EfficientNet formalized compound scaling. Each iteration improved the accuracy/cost tradeoff, but practical comparison under identical conditions is rarer than it should be.

A second concern runs parallel to raw accuracy: adversarial robustness. Dingeto & Kim (2024) showed that both CNNs and Vision Transformers (ViTs) are vulnerable to perturbation attacks — FGSM, PGD, MIM — and that regularization (specifically Elastic Net) helps more than adversarial training alone. That finding is relevant to how these benchmark models would behave outside a clean test set.

This report answers three specific questions:

- Under identical training conditions, how do five CNN architectures rank on CIFAR-10?
- What architectural factors explain the performance differences, including VGG-16's failure?
- How do these clean-accuracy results relate to adversarial robustness data from the literature?

1.1 Research Contributions

- A reproducible five-model benchmark on CIFAR-10 under strictly controlled conditions.
- A concrete explanation of the VGG-16 spatial resolution failure with architectural analysis.
- An efficiency comparison showing MobileNetV2 and EfficientNet-B0 dominate accuracy-per-parameter.
- A per-class confusion analysis — cat/dog is by far the worst boundary at 32×32 resolution.
- A cross-study table linking benchmark accuracy to adversarial robustness results from Dingeto & Kim (2024).
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2. Related Work

2.1 CNN Architecture Evolution

The VGGNet architecture demonstrated that network depth using small 3×3 convolution filters is critical for achieving strong performance, though its fully-connected layers contribute to a large parameter count (~138M for VGG-16). He et al. (2016) introduced residual connections in ResNet, enabling training of much deeper networks by mitigating the vanishing gradient problem. Sandler et al. (2018) proposed MobileNetV2, using inverted residuals and linear bottlenecks for resource-constrained deployments. Tan & Le (2019) introduced EfficientNet, which uses a compound scaling method to uniformly scale depth, width, and resolution.

2.2 Benchmarks on CIFAR-10

The CIFAR-10 dataset has been used as a benchmark since its introduction by Krizhevsky et al. (2009). Prior work has demonstrated that transfer learning from ImageNet-pretrained models significantly improves accuracy over training from scratch on small datasets. Kornblith et al. (2019) showed that ImageNet accuracy correlates strongly with transfer learning performance on downstream tasks.

2.3 Adversarial Robustness in CNNs and ViTs

Dingeto & Kim (2024) conducted a comprehensive comparative study of adversarial defenses in Vision Transformers (ViTs) and CNNs, testing models on CIFAR-10 and CIFAR-100 datasets against Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and Momentum Iterative Method (MIM) attacks. Their key finding is that regularization techniques — specifically Elastic Net (combined L1 and L2) — enable ViTs, and in most cases CNNs, to enhance adversarial defenses more effectively than traditional adversarial training, while requiring significantly fewer computational resources.

The study reveals that ViTs, owing to their global self-attention mechanism, inherently capture feature dependencies across the entire image, making them more resistant to localized adversarial perturbations. In contrast, CNNs — which process features locally through convolutional filters — remain more susceptible even with regularization applied. These findings are directly relevant to understanding the adversarial vulnerabilities of the CNN architectures benchmarked in this report.

3. Dataset and Preprocessing

3.1 CIFAR-10 Dataset

CIFAR-10 is a widely-used benchmark dataset consisting of 60,000 RGB images of size 32×32 pixels across 10 balanced classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The dataset contains exactly 6,000 images per class, partitioned into 50,000 training and 10,000 test images. The balanced class distribution justifies the use of top-1 accuracy as the primary evaluation metric without requiring class-weighted adjustments.

3.2 Normalization

All images are normalized using channel-wise mean and standard deviation computed over the entire training set: mean = (0.4914, 0.4822, 0.4465) and std = (0.2023, 0.1994, 0.2010) for R, G, B channels respectively. This zero-mean unit-variance normalization is applied to both training and test sets.

3.3 Data Augmentation

Data augmentation was applied exclusively during training to improve generalization and reduce overfitting. The following transformations are applied sequentially:

- RandomHorizontalFlip with probability 0.5, preventing mirror-image bias.
- RandomCrop of 32×32 pixels with padding of 4 pixels, providing translational invariance.
- ColorJitter with brightness=0.2 and contrast=0.2, providing robustness to lighting variation.

No augmentation was applied during evaluation.

4. Methodology

4.1 Model Architectures

Five architectures were evaluated. The SimpleCNN baseline is a three-block convolutional network trained from scratch with Conv2D(3→32), Conv2D(32→64), and Conv2D(64→128) layers, each followed by ReLU activation and MaxPool2D, with a fully-connected head (2048→512→10) and dropout regularization (p=0.5). This model contains 1.15M parameters and serves as the from-scratch training baseline.

For the four pretrained models (ResNet-18, VGG-16, MobileNetV2, EfficientNet-B0), ImageNet-pretrained weights were used and only the final classification layer was replaced with a new linear layer mapping to 10 output classes. All layers including the pretrained backbone were fine-tuned during training (full fine-tuning protocol).

4.2 Training Protocol

All models were trained under identical conditions to ensure fair comparison. CosineAnnealingLR scheduling was used rather than step decay, as it provides smoother convergence and generally superior final accuracy. All experiments were conducted on an NVIDIA Tesla T4 GPU using PyTorch 2.9.0 with CUDA 12.8.

Table 1. Hyperparameter settings applied uniformly across all model training runs.

Hyperparameter	Value	Rationale
Optimizer	Adam	Adaptive learning rate
Learning Rate	0.001	Standard Adam default
LR Scheduler	CosineAnnealingLR	Smooth convergence
Batch Size	64	GPU memory balance
Epochs	20	Convergence observed
Loss Function	CrossEntropyLoss	Multi-class standard
Dropout (Baseline)	0.5	Regularization

5. Results and Analysis

5.1 Overall Benchmark Results

Table 2 presents the complete benchmark results for all five architectures. EfficientNet-B0 achieves the highest test accuracy of 88.11%, followed closely by MobileNetV2 (87.63%) and ResNet-18 (87.33%). The SimpleCNN baseline achieves a competitive 80.87% given its limited capacity and absence of pretraining. The most notable finding is the catastrophic failure of VGG-16, which achieves only 31.06% test accuracy despite its 134.3M parameters — marginally above random chance (10%).

Table 2. Complete benchmark results. All models trained for 20 epochs under identical conditions. ★ = best accuracy.

Model	Test Acc	Train Acc	Params	Train Time	Notes
SimpleCNN (baseline)	80.87%	81.08%	1.15M	~5 min	From scratch
ResNet-18	87.33%	92.26%	11.18M	12.6 min	Pretrained
VGG-16	31.06%	10.37%	134.3M	22.7 min	FAILED — see §5.2
MobileNetV2	87.63%	90.53%	2.24M	15.2 min	Best param efficiency
EfficientNet-B0 ★	**88.11%**	93.66%	4.02M	17.8 min	Best accuracy

5.2 The VGG-16 Failure Mode

VGG-16's 31.06% is worth understanding, not just flagging. The architecture applies five max-pool operations with stride 2 — designed for ImageNet's 224×224 input. That reduces spatial dimensions by a factor of 32 in each dimension, yielding a 7×7 feature map from a 224×224 input. Fine.

Apply the same five pooling layers to CIFAR-10's 32×32 images: 32 / 32 = 1. You get a 1×1 feature map. All spatial information is gone before the classifier sees anything. The training accuracy of 10.37% — essentially random — confirms the model never learns. It's not a training instability, it's a geometry problem.

Fix: either remove pooling layers, increase input resolution via bilinear upsampling to 224×224, or adapt the architecture. Tan & Le (2019) address this explicitly — **resolution is a scaling dimension**, and ignoring it breaks the architecture.

5.3 Efficiency Analysis

Parameter count doesn't predict accuracy here. MobileNetV2 beats ResNet-18 using 5× fewer parameters (2.24M vs 11.18M). EfficientNet-B0 leads overall at 4.02M. VGG-16 uses 134.3M parameters and fails completely. The compound scaling logic in EfficientNet is doing real work — it allocates capacity efficiently across depth, width, and resolution.

SimpleCNN's 80.87% at 1.15M is the honest baseline. That's the accuracy you can get without pretrained weights, with a simple three-block architecture and good regularization. The gap to pretrained models (~7pp) is entirely explained by the prior on features.

5.4 Per-Class Analysis (EfficientNet-B0)

Truck (93.7%), automobile (92.9%), and ship (92.8%) are the easiest classes — strong structural features, low inter-class confusion. Cat is the hardest at 77.0% recall, with 9.2% misclassified as dog. Dog recall is 79.0%, with 10.7% misclassified as cat. This 9–11% bidirectional confusion is the largest error pair in the matrix.

At 32×32 resolution, cats and dogs share enough visual texture (fur, ears, facial structure) that even the best model here gets them wrong ~1 in 10 times. Higher input resolution would likely help; upsampling to 64×64 or 128×128 before inference is worth testing.

Table 3. Per-class precision, recall, F1-score and primary confusion for EfficientNet-B0. ⚠ = highest confusion pairs.

Class	Precision	Recall	F1-Score	Support	Hardest Confusion
airplane	0.89	0.92	0.90	1000	ship (2.2%)
automobile	0.94	0.93	0.94	1000	truck (3.4%)
bird	0.83	0.85	0.84	1000	deer (3.9%)
cat ⚠	0.77	0.77	0.77	1000	**dog (9.2%) ⚠**
deer	0.87	0.87	0.87	1000	bird (3.9%)
dog ⚠	0.82	0.79	0.80	1000	**cat (10.7%) ⚠**
frog	0.91	0.91	0.91	1000	cat (2.5%)
horse	0.91	0.91	0.91	1000	dog (3.1%)
ship	0.94	0.93	0.94	1000	airplane (3.6%)
truck	0.91	0.94	0.93	1000	automobile (2.9%)

6. Discussion

6.1 Transfer Learning vs. Training from Scratch

The consistent gap between the SimpleCNN baseline (80.87%) and all successfully pretrained models (87–88%) confirms the value of transfer learning even for architectures pretrained on a different domain (ImageNet vs. CIFAR-10). This gap of approximately 7 percentage points is achieved without any architecture modifications, purely through the use of ImageNet-pretrained initialization. This is consistent with Kornblith et al. (2019), who demonstrate that better ImageNet models transfer better to downstream tasks.

6.2 CNN Benchmark vs. Adversarial Robustness

The benchmark results become particularly meaningful when contextualized against adversarial robustness findings. Dingeto & Kim (2024) evaluated similar CNN architectures on CIFAR-10 under adversarial attacks (FGSM, PGD, MIM) and found that a regularized CNN achieved 74.7% clean accuracy and 73.8% average adversarial accuracy — comparable to the SimpleCNN baseline of 80.87% under clean conditions in this benchmark.

Crucially, the same study found that Vision Transformers with Elastic Net regularization achieved 84.0% clean accuracy and 82.2% adversarial accuracy on CIFAR-10, outperforming even EfficientNet-B0's benchmark accuracy of 88.11% in adversarial scenarios. This suggests that while EfficientNet-B0 leads purely on clean data accuracy, a regularized ViT would likely prove more resilient in deployment environments where adversarial manipulation is a concern.

The table below summarizes this cross-study comparison for CIFAR-10:

Table 4. Cross-study comparison of CNN benchmark results vs. adversarial robustness literature on CIFAR-10.

Model	Clean Accuracy	Adv. Accuracy	Source
EfficientNet-B0	88.11%	—	This benchmark
MobileNetV2	87.63%	—	This benchmark
ResNet-18	87.33%	—	This benchmark
Regularized ViT (Elastic Net)	**84.0%**	**82.2%**	**Dingeto & Kim (2024)**
Regularized CNN (Elastic Net)	74.7%	73.8%	Dingeto & Kim (2024)
SimpleCNN (Baseline)	80.87%	—	This benchmark

6.3 Practical Model Selection Guidance

For practitioners seeking to deploy image classifiers in resource-constrained environments, this benchmark provides the following evidence-based guidance:

- MobileNetV2 offers the best accuracy-to-parameter ratio: 87.63% at 2.24M parameters with the shortest training time (15.2 min) among pretrained models.
- EfficientNet-B0 provides the highest absolute clean-data accuracy (88.11%) at modest cost (4.02M parameters).
- ResNet-18 is a reliable choice when compatibility with standard deep learning infrastructure is important.
- VGG-16 should be avoided for small-resolution datasets unless the architecture is explicitly adapted (removing pooling layers or upsampling inputs).
- For adversarially sensitive deployments, regularized ViTs (Elastic Net) should be preferred, as they achieve competitive accuracy with substantially greater robustness against FGSM, PGD, and MIM attacks.

6.4 Limitations

Several limitations of this study should be acknowledged. First, all models were trained for a fixed 20 epochs; additional training may improve results, particularly for larger models. Second, the benchmark is limited to CIFAR-10; different conclusions may hold for other datasets with different resolutions or class distributions. Third, inference time and memory footprint — important practical metrics — were not measured. Fourth, the adversarial robustness comparison is cross-study and does not involve direct adversarial testing of the benchmarked architectures under identical conditions.

7. Conclusion

This report presented a systematic benchmark of five CNN architectures on CIFAR-10 under identical training conditions. EfficientNet-B0 achieved the best test accuracy of 88.11% with only 4.02M parameters, followed closely by MobileNetV2 (87.63%, 2.24M) and ResNet-18 (87.33%, 11.18M). VGG-16 failed catastrophically due to spatial resolution collapse caused by its pooling architecture — a finding with important implications for architecture selection on small-image datasets.

The results demonstrate that modern efficient architectures (MobileNetV2, EfficientNet-B0) offer superior accuracy-per-parameter trade-offs compared to older, larger architectures. Transfer learning provides a consistent ~7% accuracy improvement over training from scratch. The cat/dog confusion pair (9–11%) represents the primary error mode of the best-performing model.

Critically, contextualizing these benchmark results against adversarial robustness literature reveals that while EfficientNet-B0 leads on clean data, regularized Vision Transformers (using Elastic Net regularization) achieve greater adversarial resilience at competitive accuracy levels. This synthesis suggests that future work should combine the architectural efficiency insights from CNN benchmarking with regularization-based robustness strategies from ViT research, particularly for security-sensitive real-world deployments.

Future directions include extending this benchmark to CIFAR-100 and higher-resolution datasets, incorporating inference speed and memory benchmarks, evaluating the impact of longer training schedules, and directly testing benchmarked CNN architectures under adversarial attack conditions.

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Appendix A: Reproducibility Notes

All experiments were conducted on Google Colab using an NVIDIA Tesla T4 GPU. The complete source code, model checkpoints, and training logs are available at: https://github.com/MRNAYAK-DA/cnn-cifar10-benchmark. The following software versions were used: PyTorch 2.9.0, torchvision 0.20.0, CUDA 12.8, Python 3.10. All random seeds were not fixed in this study; minor accuracy variations (±0.5%) across runs are expected.

Appendix B: Training Convergence Notes

Training and validation loss curves demonstrate healthy convergence for all models except VGG-16. For SimpleCNN, test accuracy exceeded training accuracy in early epochs (1–10), attributed to training augmentation making training harder than evaluation — a sign of correctly implemented augmentation. For all pretrained models, training accuracy plateaued near 90–94% by epoch 15–20, while test accuracy stabilized at 87–88%, indicating slight overfitting that could be addressed with longer training or stronger regularization.